

Breathing Training and Acute Breath-Regulation Strategies in Sport and Performance Contexts: A Scoping Review

Review authors

June 19, 2026

Abstract

Background: Breath-regulation practices are increasingly used in sport, exercise, and performance settings, but the evidence base spans heterogeneous interventions, populations, outcomes, and study designs. **Objective:** This scoping review mapped the available evidence on breathing-training and acute breath-regulation strategies in athletic or performance-relevant contexts, summarized study characteristics, charted intervention categories and outcomes, and identified evidence gaps for future trials and reviews. **Eligibility criteria:** Sources were eligible when they examined a breathing intervention or breathing-related exposure relevant to sport, exercise, or performance, including hyperventilation, hypoventilation, paced breathing, pranayama/yogic breathing, breath holding, hook breathing, Wim Hof-type breathing, or closely related approaches. **Sources of evidence:** The review started from 272 PubMed records, retained 43 after title/abstract screening, retained 31 after full-text screening, and recovered three additional papers by citation backtracking. **Charting and synthesis:** Data were charted into bibliometric, demographic, intervention, outcome, feasibility, and safety tables, then summarized descriptively following PRISMA-ScR guidance. Where sufficiently comparable outcome clusters existed, exploratory standardized mean differences were calculated. **Results:** Thirty-four sources were included. Most were experimental (31/34, 91.2%), but many were crossover, single-session, small-sample, or otherwise classified as experimental-other. Evidence was concentrated in hyperventilation, paced breathing, hypoventilation, pranayama/yogic breathing, and breath-holding strategies, with broad variation in sport context and outcome selection. Exploratory pooling suggested a small, borderline effect favoring hyperventilation for power outcomes ($g = 0.30$, 95% CI -0.00 to 0.60), while paced-breathing effects on reaction outcomes were inconclusive ($g = -0.15$, 95% CI -0.45 to 0.15). Feasibility reporting was limited, and adherence and dropout were rarely reported in a form suitable for cross-study synthesis. **Conclusions:** The field is active but fragmented. Current evidence is most useful for identifying promising mechanisms and study targets rather than making definitive practice recommendations. Future work should use preregistered protocols, clearer intervention taxonomies, sport-specific outcomes, adverse-event reporting, and adequately powered comparative designs.

Key takeaways in plain language

- Breathing practices in sport look promising, but the evidence is still scattered across many techniques, sports, and outcomes.
- Slow-paced breathing, breath holding, hypoventilation, and hyperventilation should not be treated as the same intervention.
- Hyperventilation showed a small signal for power outcomes, while paced breathing did not show a clear reaction-time benefit in the available pooled data.
- Future studies should test sport-developed breathing routines more rigorously and report adherence, dropout, and adverse events.

Keywords: scoping review; PRISMA-ScR; breathing training; breathwork; hyperventilation; hypoventilation; paced breathing; sport performance; athletes

1 Introduction

Optimal respiratory strategies represent a physiological mechanism capable of modulating autonomic nervous-system activity, psychological stress, arousal regulation, and physical execution in competitive sport (Migliaccio et al., 2023). Breathing is therefore both an automatic physiological process and a deliberately modifiable behavior. In sport and exercise settings, athletes and practitioners use breath-regulation strategies to influence acid-base balance, tolerance to hypoxia or hypercapnia, respiratory muscle function, perceived exertion, recovery, attention, and emotional control.

Despite increasing integration into athletic training, the categorization and validation of breathing techniques performed without external devices remain fragmented and lack standardized empirical protocols (Laborde et al., 2024). The term “breathing technique” is often used as if it described a single intervention class, but the practices grouped under that label are mechanistically diverse. The included studies ranged from acute hyperventilation or Wim Hof-type breathing to repeated-sprint hypoventilation, slow-paced or coherent breathing, box breathing, pranayama, Bhramari breathing, breath holding, Valsalva-type strategies, hook breathing in freediving, and observational comparisons of breathing patterns.

This heterogeneity matters for interpretation. Hyperventilation may acutely alter carbon dioxide, pH, ventilatory discomfort, and repeated high-intensity exercise tolerance. Hypoventilation or breath-holding strategies may expose athletes to hypoxemic or hypercapnic stress. Paced breathing is more often framed as an autonomic, attentional, or anxiety-regulation strategy. Pranayama and related yogic methods combine respiratory control with broader psychophysiological training. A systematic evaluation of the literature is therefore needed to consolidate empirical evidence, clarify how breathing techniques are being categorized, and evaluate their effects on sport performance.

This heterogeneity also makes a scoping review more appropriate than a narrowly framed systematic review of efficacy. Scoping reviews are designed to map a body of literature, clarify concepts, identify types of available evidence, and locate gaps where later systematic reviews or trials may be justified (Arksey & O'Malley, 2005; Levac et al., 2010; Peters et al., 2020). Reporting followed the PRISMA extension

for scoping reviews (PRISMA-ScR), which emphasizes transparent reporting of eligibility criteria, evidence sources, selection, data charting, and synthesis (Tricco et al., 2018).

The primary objectives of this scoping review were to map the literature on breathing techniques executed without external devices and evaluate the available evidence on their effects on sports performance. Specifically, we aimed to describe publication trends, study designs, populations, breathing-intervention categories, dosing characteristics, outcome domains, feasibility and safety reporting, and any comparable quantitative outcome clusters emerging from the extracted material.

2 Methods

2.1 Design and reporting framework

This scoping review was reported according to PRISMA-ScR (Tricco et al., 2018). The review was designed to map the range, characteristics, and reporting quality of studies examining non-device breathing strategies in sport, exercise, and performance-relevant settings. No protocol was registered.

2.2 Eligibility criteria

Sources were considered eligible when they examined a breathing intervention, breathing maneuver, or breathing-related exposure relevant to sport, exercise, or performance. Eligible concepts included hyperventilation, hypoventilation, paced or coherent breathing, box breathing, pranayama or yogic breathing, breath holding, Valsalva-type strategies, hook breathing, Wim Hof-type breathing, diaphragmatic breathing, and similar practices. Eligible contexts included athletes, trained or recreationally active participants, sport-specific tasks, exercise performance, recovery, psychophysiological preparation, and laboratory performance tasks with plausible sport relevance. No formal restrictions on study design were applied during charting.

2.3 Information sources and selection

The review began with a PubMed search performed on 15 June 2025, which returned 272 records. The search strategy was:

```
((("Healthy Volunteers"[Mesh] OR "Athletes"[Mesh] OR healthy[Title/Abstract] OR athlete*[Title/Abstract] OR sport*[Title/Abstract])) AND ("breath-holding"[Title/Abstract] OR "apnea"[Title/Abstract] OR "breathing exercise"[Title/Abstract] OR "controlled breathing"[Title/Abstract] OR "paced breathing"[Title/Abstract] OR "diaphragmatic breathing"[Title/Abstract] OR "hyperventilation"[Title/Abstract] OR "pranayama"[Title/Abstract] OR "Wim Hof"[Title/Abstract])) AND ("performance"[Title/Abstract] OR "endurance"[Title/Abstract] OR "sports performance"[Title/Abstract] OR "competition"[Title/Abstract])) NOT ("review"[Publication Type] OR "Respiratory Muscle Training"[Title/Abstract] OR "Inspiratory Muscle Training"[Title/Abstract] OR "COPD"[Title/Abstract] OR "asthma"[Title/Abstract] OR "sleep apnea"[Title/Abstract] OR "stroke"[Title/Abstract] OR "pulmonary disease"[Title/Abstract] OR "dyspnea"[Title/Abstract])
```

After title and abstract screening, 43 records were retained for full-text assessment. Thirty-one reports were retained after full-text screening. Citation backtracking recovered three additional reports: Taylor 2021, Fernandez 2019, and Fernandez 2025. The final evidence base therefore contained 34 sources.

2.4 Data charting

Data were charted into six domains: bibliographic information, descriptive characteristics, intervention specifics, intervention outcomes, control or comparator outcomes, and feasibility/safety. Charted variables included publication year, study design, sport discipline, athlete level, sample size, age, sex distribution, breathing classification, dose frequency, dose duration, comparator type, outcome metric, group sample sizes, group means and dispersions, effect estimates, adverse events, adherence, dropout, and feasibility scores.

2.5 Synthesis

The primary synthesis was descriptive and evidence-mapping oriented. Publication year, study design, sport context, participant characteristics, breathing categories, dose frequency, dose duration, outcome clusters, and feasibility/safety fields were summarized using frequencies, percentages, medians, interquartile ranges, and narrative synthesis as appropriate. For outcome clusters with at least two sufficiently comparable complete records, exploratory Hedges' g values and random-effects models were calculated. These analyses were treated as exploratory adjuncts to the scoping review rather than as definitive meta-analytic evidence.

2.6 Critical appraisal

Consistent with PRISMA-ScR, formal risk-of-bias appraisal was not undertaken because the central objective was to map the available evidence rather than estimate a single intervention effect. Completeness of reporting, comparability of outcomes, and feasibility/safety reporting were considered when interpreting the strength and applicability of the evidence.

3 Results

3.1 Selection of sources

The selection process is shown in Figure 1. From 272 PubMed records, 43 were retained after title and abstract screening. Thirty-one were included after full-text screening, and three additional papers were recovered by citation backtracking, yielding 34 included sources.

Figure 1 shows a large reduction during title/abstract screening, with 229 records excluded before full-text assessment. The final corpus combines the 31 full-text inclusions with three citation-backtracked papers, which is important because the backtracked studies contributed directly to the final map rather than serving only as contextual references.

3.2 Study and Participant Characteristics

Table 1: Baseline study and participant characteristics from tables/table1.csv.

Characteristic	Summary	Denominator / missing data
Included studies	34 studies	34/34 reported
Aggregate sample size	1,593 participants	Reported in 30/34 studies; 4 unreported
Study sample size	Mean $\pm$ SD: 53.1 $\pm$ 97.9; Median [IQR]: 29.0 [19.3–46.8]	Calculated among 30 studies reporting N_Total
Study mean age	Mean $\pm$ SD: 22.3 $\pm$ 6.1 years; Median [IQR]: 21.3 [19.8–23.7]	Calculated among 26 studies reporting Age_Mean; 8 unreported
Estimated sex distribution	Male: 827.0 (53.5%); Female: 720.0 (46.5%)	Estimated among 1,547 participants from 29 studies reporting both sample size and male percentage; 5 studies lacked sufficient data
Athlete level	Competitive: 14 (41.2%); Recreational: 12 (35.3%); Elite: 8 (23.5%)	34/34 studies classified
Study design	Experimental other: 19 (55.9%); Experimental RCT: 9 (26.5%); Observational cross-sectional: 3 (8.8%); Experimental before-after: 2 (5.9%); Observational retrospective: 1 (2.9%)	34/34 studies classified
Sport discipline	Not sport-specific/none: 7 (20.6%); Various: 5 (14.7%); Running: 4 (11.8%); Swimming: 3 (8.8%); Freediving: 3 (8.8%); Resistance training: 2 (5.9%); single-study disciplines: 10 (29.4%)	34/34 studies classified
Race or ethnicity reporting	Reported: 0 (0.0%); Not reported: 34 (100.0%)	34/34 studies assessed
Baseline anthropometrics reported	Reported: 23 (67.6%); Not reported: 11 (32.4%)	34/34 studies assessed
Attrition noted	Yes: 8 (23.5%); No: 26 (76.5%)	34/34 studies assessed

Table 1 shows that the mapped evidence base primarily reflects young adult cohorts, with sample-size information available for most studies and an aggregate sample exceeding 1,500 participants. The participant pool was modestly male-predominant among studies reporting sex data, and the cohort base was mainly competitive or recreational rather than exclusively elite. Reporting was most consistent for design and athlete level, and less consistent for age, race/ethnicity, anthropometric variables, and attrition.

Table 2 gives the compact profile of the included evidence base and shows why the review is better read as a map than as a single efficacy estimate. Table 3 separates the main breathing-technique categories, making clear that “breathwork” covers interventions with different physiological targets. Table 4

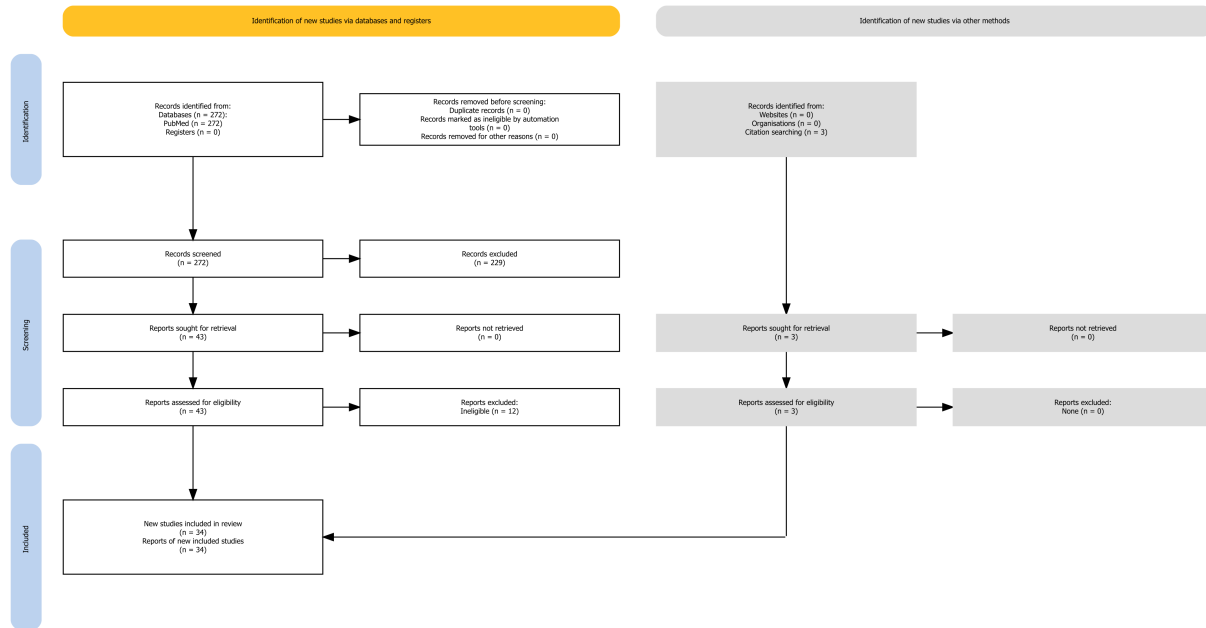


Figure 1: PRISMA 2020 source selection flow.

Table 2: Summary of the mapped evidence base.

Domain	Finding
Initial PubMed records	272
Title/abstract retained	43
Full-text retained	31
Citation-backtracked additions	3
Included sources in extraction dataset	34
Publication years represented	2004–2026 in cleaned extraction; one source has a 2000 bibliography year
Experimental studies	31/34 (91.2%)
Observational studies	2/34 (5.9%)
Unclear design classification	1/34 (2.9%)
Studies with age data	26 studies, 1185 participants
Weighted mean age	26.7 years (pooled SD 10.2)
Studies with sex data	1234 participants; estimated male proportion 51.5%

Table 3: Breathing-training categories in the extracted intervention records.

Category	Records	Percentage
Other or unclear	11	31.4%
Hyperventilation	9	25.7%
Paced breathing	9	25.7%
Hypoventilation	3	8.6%
Pranayama/yogic breathing	2	5.7%
Breath holding/Valsalva	1	2.9%

Table 4: Most represented sport contexts.

Sport context	Studies	Percentage
Not sport-specific	7	20.6%
Various/mixed sports	6	17.6%
Freediving	3	8.8%
Running	3	8.8%
Swimming	3	8.8%
Resistance training	2	5.9%
Single-study contexts	10	29.4%

Table 5: Exploratory random-effects syntheses from comparable outcome clusters.

Cluster	k	Pooled g	95% CI	p	I^2
Hyperventilation and power outcomes	6	0.30	-0.00 to 0.60	0.052	0.0%
Paced breathing and reaction outcomes	4	-0.15	-0.45 to 0.15	0.335	57.9%
Hypoventilation and repeated-sprint outcomes	1	–	Not pooled	–	–

Positive values were coded as favoring the breathing intervention after direction normalization. These syntheses are exploratory because outcome clusters were small and heterogeneous.

Table 6: Feasibility and safety reporting.

Item	Extracted finding
Studies with parseable adherence data	0
Studies with parseable dropout data	0
Adverse event mentions	Dizziness, heaviness, numbness, tingling, and pulmonary edema
Main safety pattern	Hyperventilation-related sensory symptoms were reported in one source; pulmonary edema was noted in a freediving/hook-breathing context

shows that the literature is distributed across several sport contexts rather than concentrated in one mature performance domain. Table 5 reports the two exploratory quantitative clusters that were sufficiently comparable for pooling. Table 6 highlights the weakest reporting area: adherence, dropout, feasibility, and safety were not documented consistently enough to support strong implementation guidance.

3.3 Characteristics of included sources

The evidence map included 34 sources spanning 2004–2026. Most studies were experimental (31/34, 91.2%), while two were observational and one remained unclear. However, only eight records were classified as parallel or controlled trials; most were crossover, repeated-measures, acute, pilot, or otherwise categorized as experimental-other.

Across studies with age data, 1185 participants contributed to a weighted mean age of 26.7 years. Sex data were available for 1234 participants and suggested an approximately balanced overall sample, with an estimated male proportion of 51.5%. Sport contexts were dispersed. The most frequent categories were not sport-specific, mixed/various sports, freediving, running, and swimming, while many sport contexts appeared only once.

Table 7: Characteristics of included sources.

Study ID	Year	Design	Sport/context	Level/population	Breathing category
Durand2004	2004	Observational cross-sectional	Endurance	Competitive	Other/unclear
Spiering2004	2004	Experimental, other	Various	Competitive	Other/unclear
Lindholm2005	2005	Experimental, other	Freediving	Elite	Other/unclear
Malakhov2014	2014	Experimental, other	Various	Competitive	Hyperventilation
Sakamoto2014	2014	Experimental, other	Various	Competitive	Hyperventilation
Sakamoto2015	2015	Experimental, other	Various	Competitive	Hyperventilation
Stavrou2015	2015	Experimental RCT	Swimming	Competitive	Hypoventilation
Hakked2017	2017	Experimental RCT	Swimming	Competitive	Pranayama/yogic
Sakamoto2018	2018	Experimental, other	Various	Competitive	Hyperventilation
Fernandez2019	2019	Experimental, other	Freediving	Elite	Other/unclear
Citherlet2021	2021	Experimental, other	Running	Recreational	Hyperventilation; Wim Hof
Taylor2021	2021	Experimental, other	Not sport-specific	Recreational	Paced breathing
Vaidya2021	2021	Experimental, other	Cricket	Competitive	Pranayama/yogic
Bahensky2021	2021	Experimental RCT	Running	Competitive	Pranayama/yogic
Marko2022	2022	Experimental RCT	Running	Elite	Hyperventilation; Wim Hof
You2023	2023	Experimental, other	Not sport-specific	Recreational	Paced breathing
Braham2024	2024	Experimental RCT	Soccer	Competitive	Hypoventilation
Li2024	2024	Experimental, other	Not sport-specific	Recreational	Paced breathing
Merlin2024	2024	Experimental RCT	Swimming	Elite	Paced breathing
Sikora2024	2024	Observational cross-sectional	Various	Competitive	Other/unclear
Woorons2024	2024	Experimental RCT	Judo	Elite	Hypoventilation

Study ID	Year	Design	Sport/context	Level/population	Breathing category
Deniz2025	2025	Experimental, other	Resistance training	Recreational	Breath holding; Valsalva
Fernandez2025	2025	Experimental, other	Freediving	Elite	Other/unclear
Fesseler2025	2025	Experimental, other	Running	Recreational	Hyperventilation
Fox2025	2025	Experimental RCT	Not sport-specific	Recreational	Hyperventilation; Wim Hof
Kasap2025	2025	Experimental, other	Cycling	Recreational	Paced breathing
Yilmaz2025	2025	Experimental, other	Team sports	Competitive	Hyperventilation
Chiron2026	2026	Experimental RCT	Track and field	Elite	Paced breathing
Devipriya2026	2026	Experimental before-after	Not sport-specific	Recreational	Pranayama/yogic
FernandezBarradas2026	2026	Experimental before-after	American football	Competitive	Paced breathing
Iskra2026	2026	Experimental, other	Not sport-specific	Recreational	Paced breathing
Katz2026	2026	Experimental, other	Resistance training	Recreational	Other/unclear
Raidl2026	2026	Experimental, other	Not sport-specific	Recreational	Paced breathing
Tomita2026	2026	Experimental, other	Skating	Competitive	Paced breathing

Table 7 provides the study-level detail behind these summaries. It shows that the map includes elite, competitive, and recreational cohorts, but also that many sports and techniques are represented by only one or two studies, limiting direct sport-by-sport inference.

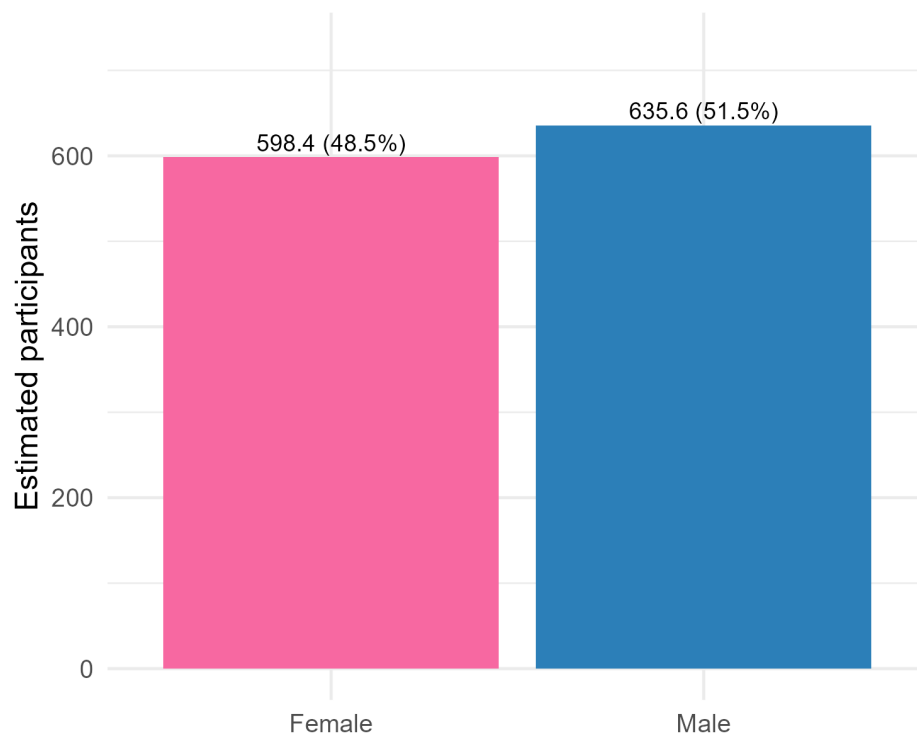


Figure 2: Estimated sex distribution among studies reporting sex data.

Figure 2 indicates that the extractable participant pool was approximately balanced overall, but this aggregate masks study-level imbalances. Several sport-specific subfields, such as freediving, resistance training, and some team-sport contexts, were male-dominant in the extraction, while other studies contained mixed or female samples. This limits certainty about whether effects generalize equally across sex and sport contexts.

3.4 Publication trends

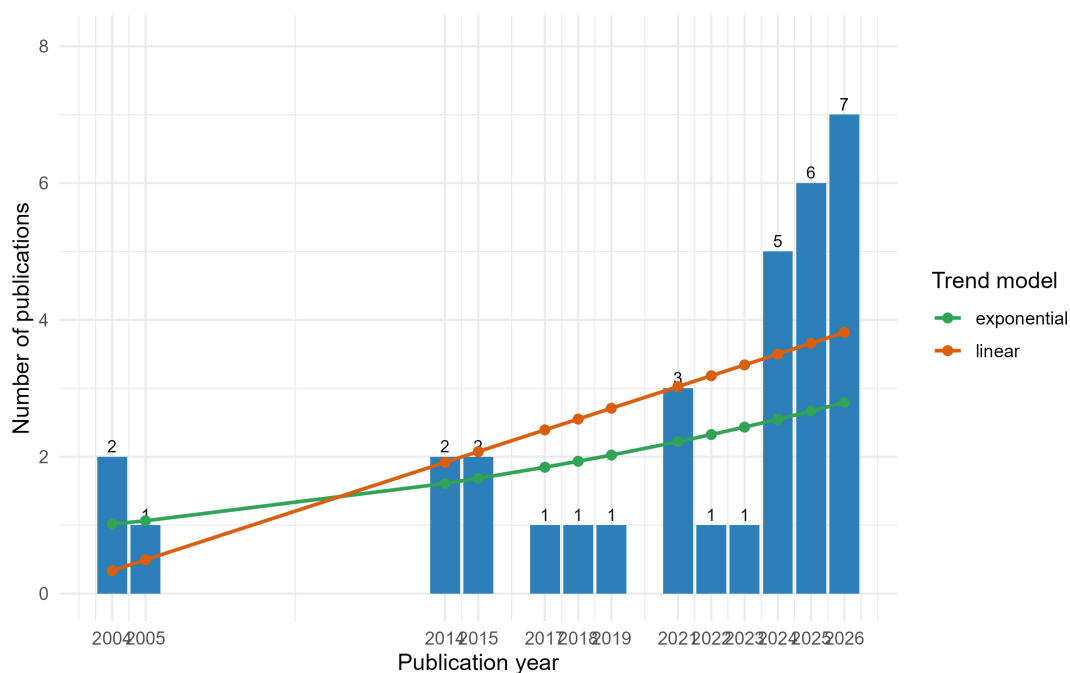


Figure 3: Annual publication counts and fitted exploratory trend lines.

Figure 3 shows that publication volume increased most clearly in the most recent years, with the highest annual counts in 2025 and 2026. Two exploratory trend models were fitted to the annual counts. A linear model estimated a slope of $+0.16$ publications per year ($R^2 = 0.28$, $p = 0.064$). An exponential model estimated an annual growth rate of 0.046 (approximately $+4.7\%$ per year; $R^2 = 0.19$, $p = 0.139$). Neither model reached conventional statistical significance given the short and irregular series, but the exponential model provided a better fit by both AIC and BIC, consistent with accelerating rather than linear accumulation. The pattern suggests growing interest in non-device breathing techniques for sport and performance, while remaining best interpreted as a descriptive trend rather than a formal forecast.

3.5 Study design and sport context

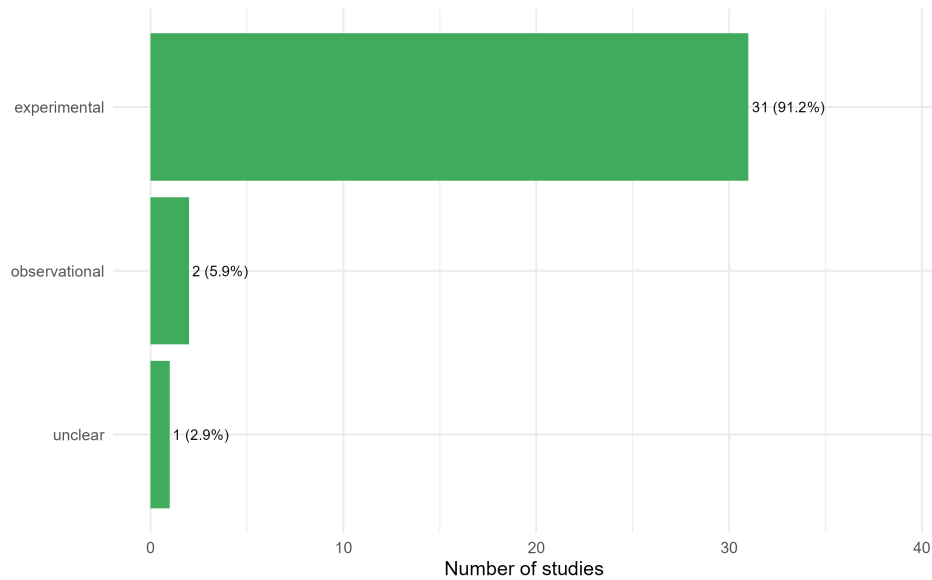


Figure 4: Broad study-design distribution.

Figure 4 shows that the evidence base is dominated by experimental studies. This is encouraging for intervention mapping, but the category is broad and includes acute crossover, repeated-measures, pilot, and before-after designs as well as randomized controlled trials.

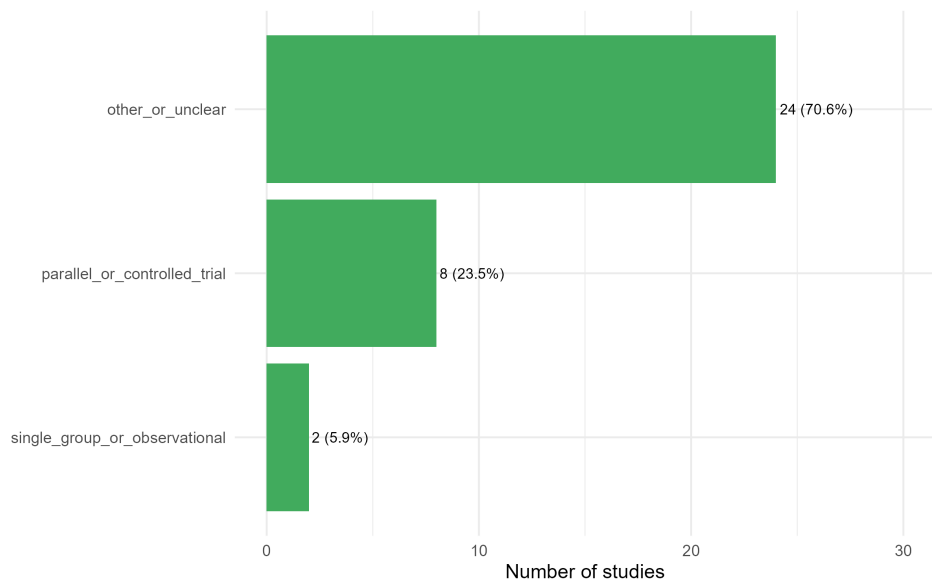


Figure 5: Detailed study-design distribution after cleaning.

Figure 5 clarifies that most records were categorized as other or unclear experimental designs, while parallel or controlled trials represented a smaller portion of the map. This supports the main methodological conclusion: the literature is active, but many studies are not yet designed or reported in a way that supports strong causal or sport-practice recommendations.

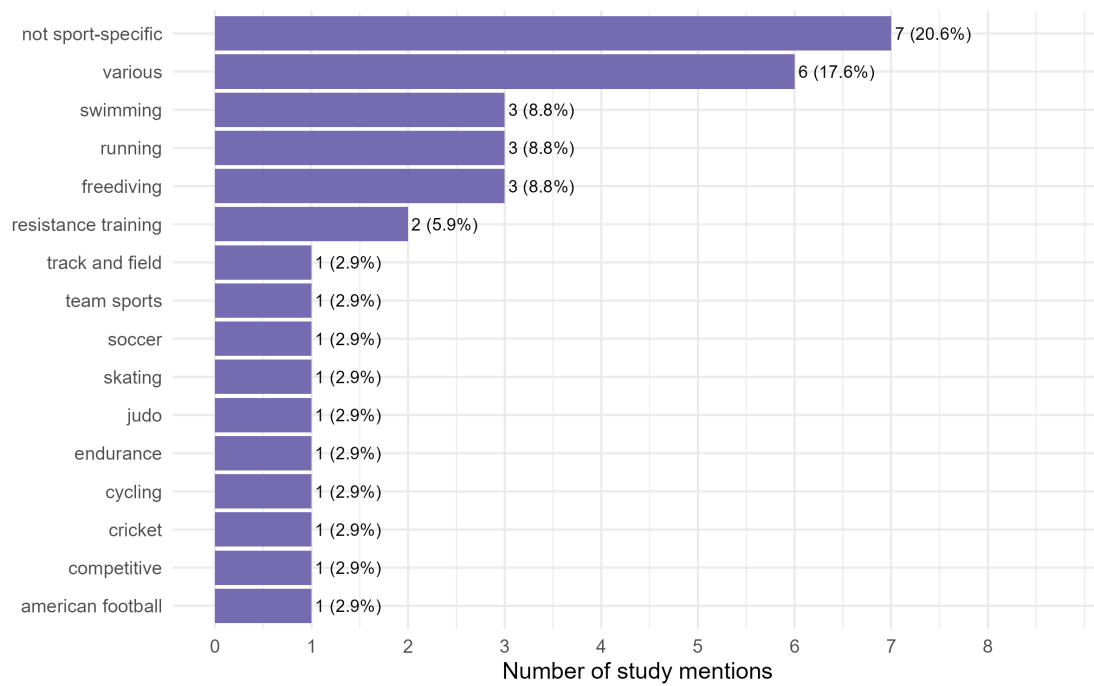


Figure 6: Distribution of sport and performance contexts.

Figure 6 shows that sport context was widely dispersed. Not sport-specific and mixed-sport studies formed the largest groups, followed by freediving, running, and swimming. The long tail of single-study sport contexts means that many practical sport-specific questions cannot yet be answered from replicated evidence.

3.6 Breathing-intervention categories and dose

The most frequent extracted categories were other/unclear breathing strategies, hyperventilation, and paced breathing. Hypoventilation, pranayama/yogic breathing, and breath holding or Valsalva-type strategies were less common. Dose reporting was uneven. Among parseable records, the median frequency was one session per week, and the median duration was one day, indicating that much of the literature concerns acute or short exposure protocols rather than extended training programs.

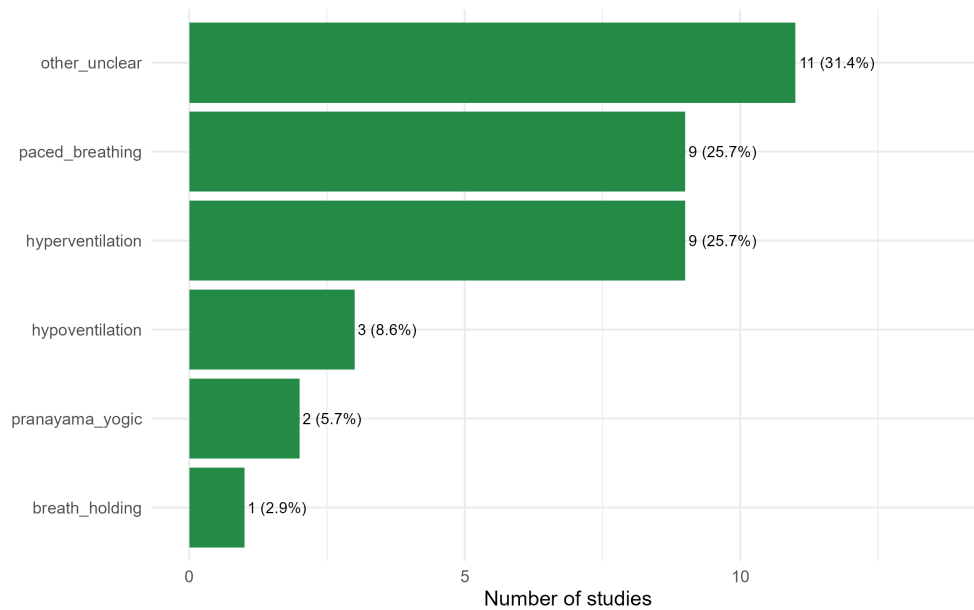


Figure 7: Distribution of breathing-training categories among included studies.

Figure 7 shows that the literature is not organized around one dominant intervention. Hyperventilation and paced breathing were equally represented among classified categories, whereas hypoventilation, pranayama/yogic breathing, and breath holding were less frequent. The large other/unclear category reflects both genuinely mixed interventions and inconsistent reporting terminology.

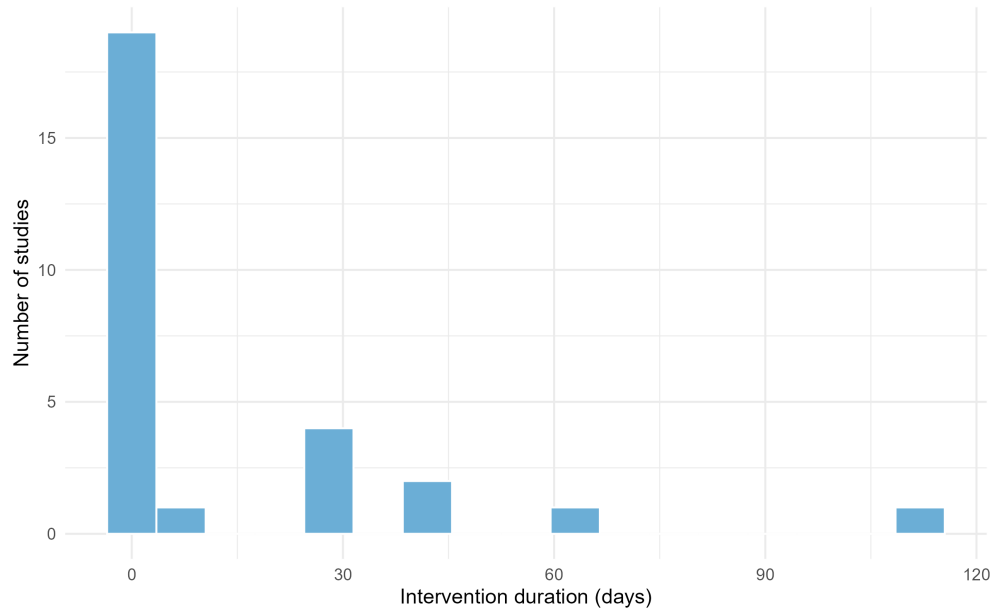


Figure 8: Distribution of intervention duration among studies reporting duration.

Figure 8 reinforces that the mapped evidence is heavily weighted toward acute or very short protocols. A smaller subset used multi-week interventions, which makes it difficult to separate immediate psychophysiological effects from adaptations that require repeated training exposure.

3.7 Outcome synthesis

Outcome measures were highly heterogeneous, covering pulmonary function, blood gases, sprint and power outcomes, reaction and movement time, heart-rate variability, perceived exertion, anxiety, recovery states, oxygen saturation, and sport-specific performance. Because most outcome domains were represented by few studies, the scoping synthesis emphasizes mapping rather than definitive effect estimation.

Exploratory quantitative synthesis was possible for two outcome clusters. Hyperventilation-related power outcomes showed a small pooled effect favoring the intervention ($g = 0.30$, 95% CI -0.00 to 0.60, $p = 0.052$, $I^2 = 0.0\%$). This estimate is compatible with a small beneficial effect but is borderline and should not be read as conclusive. Paced-breathing reaction outcomes were inconclusive ($g = -0.15$, 95% CI -0.45 to 0.15, $p = 0.335$, $I^2 = 57.9\%$). A hypoventilation repeated-sprint cluster had only one eligible effect and was not pooled.

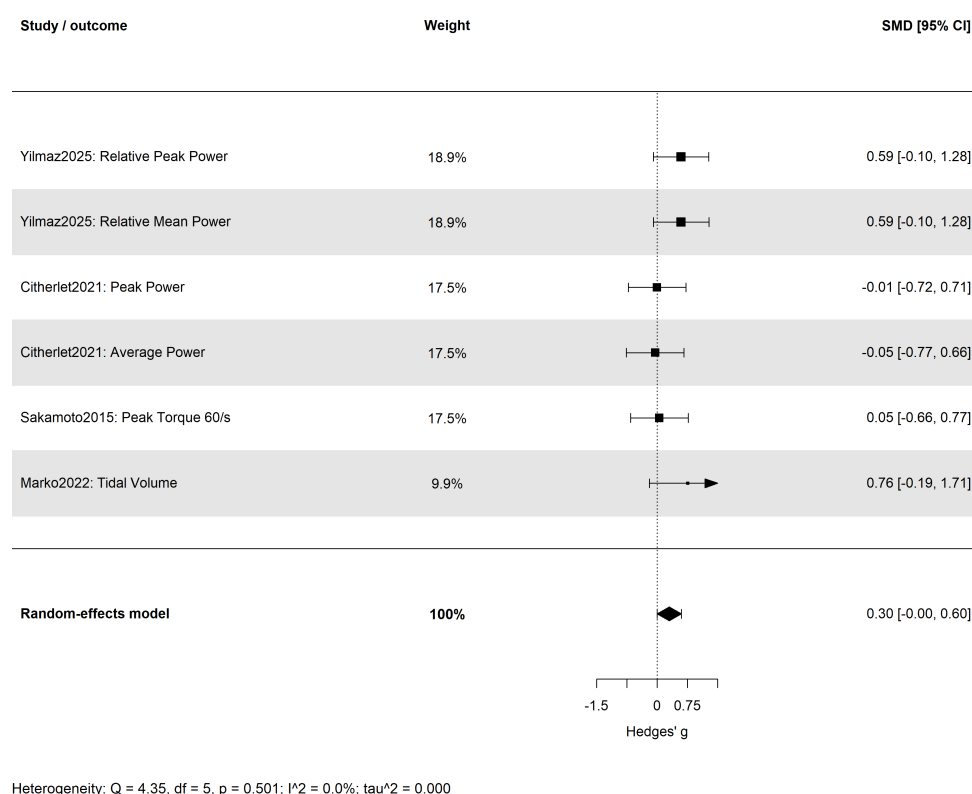


Figure 9: Exploratory forest plot for hyperventilation and power-related outcomes.

Figure 9 shows a small pooled effect close to conventional statistical significance. Individual effects were not uniformly large, but the low estimated heterogeneity indicates that this may be one of the more coherent outcome clusters in the current map. Even so, the confidence interval touches the null, so the result should be treated as a hypothesis for targeted trials rather than proof of ergogenic benefit.

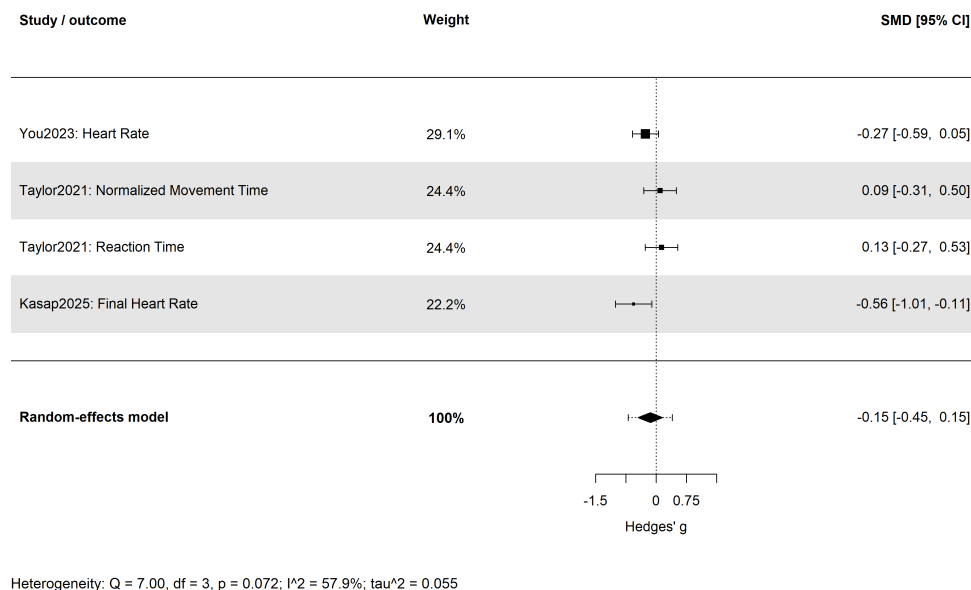


Figure 10: Exploratory forest plot for paced breathing and reaction-related outcomes.

Figure 10 does not show a clear pooled benefit for reaction-related outcomes, and heterogeneity was moderate. This suggests that paced breathing may be context-dependent: effects may depend on whether the target is arousal control, recovery, anxiety, attentional readiness, or immediate motor response.

3.8 Feasibility and safety

Feasibility reporting was sparse. Adherence and dropout were rarely reported in a form that could be summarized across studies. Adverse-event extraction identified sensory symptoms associated with hyperventilation, including dizziness, heaviness, numbness, and tingling, and pulmonary edema in a freediving/hook-breathing context. These findings do not establish incidence rates because reporting was incomplete and inconsistent, but they indicate that safety reporting should be mandatory in future studies of respiratory maneuvers that alter carbon dioxide, oxygen saturation, intrathoracic pressure, or autonomic state.

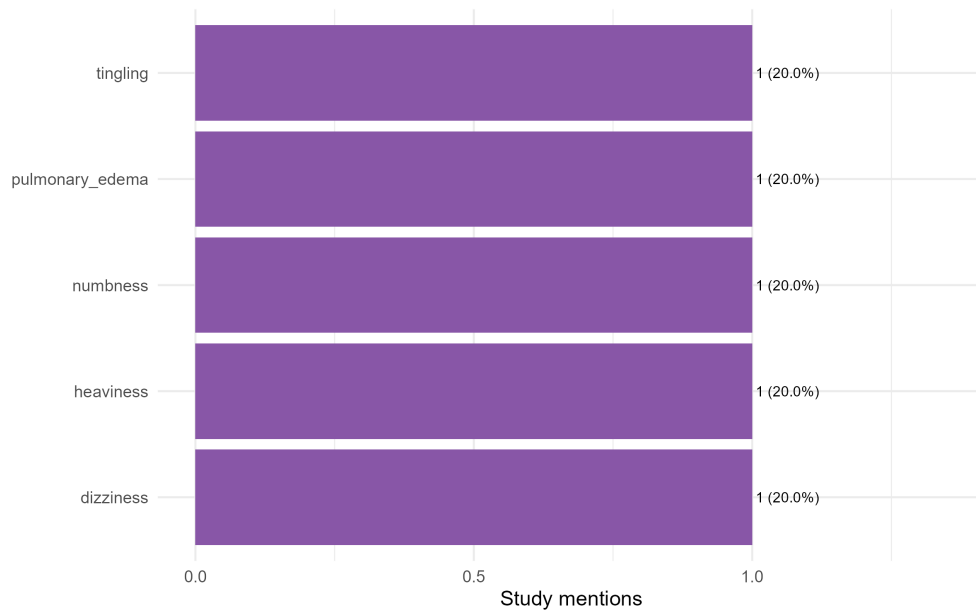


Figure 11: Adverse-event mentions by breathing-technique category.

Figure 11 shows few reported events, but the pattern is clinically relevant. Hyperventilation-related sensory symptoms are plausible consequences of hypocapnia and respiratory discomfort, while pulmonary edema in a freediving context points to a higher-risk domain where breathing maneuvers interact with pressure, hypoxia, and post-dive recovery.

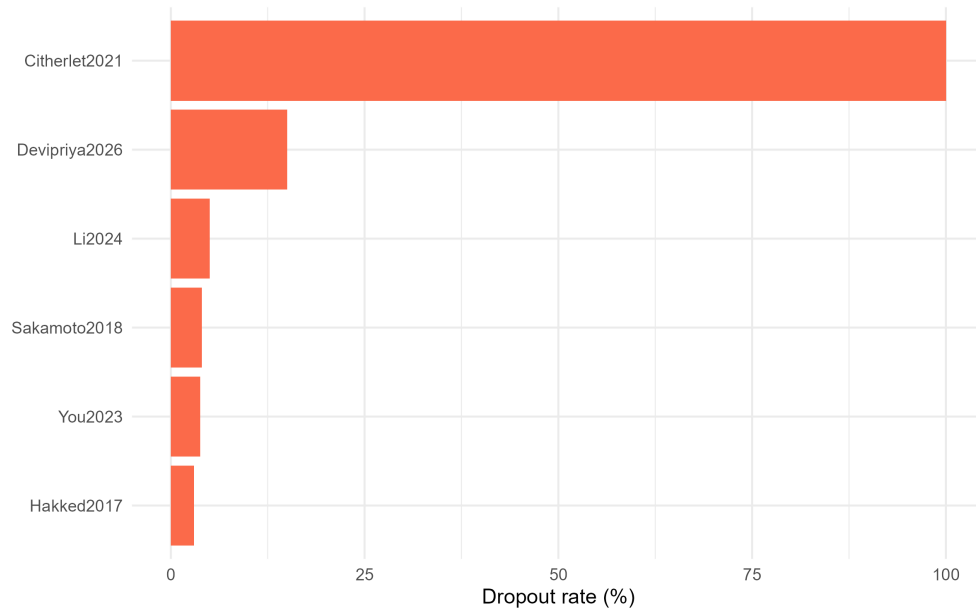


Figure 12: Dropout reporting across included studies.

Figure 12 illustrates the limited availability of consistently reported dropout data. The practical interpretation is therefore not that dropout was absent, but that dropout reporting was not extractable in a standardized way from the charted records.

4 Discussion

4.1 Summary of evidence

This scoping review mapped 34 sources on breathing-training and breath-regulation strategies in sport, exercise, and performance-relevant contexts. The evidence base is broad but fragmented. The included sources span several mechanistically different practices, including hyperventilation, hypoventilation, paced breathing, pranayama/yogic breathing, breath holding, Valsalva-type strategies, hook breathing, and Wim Hof-type breathing. Most studies were experimental, but the dominant pattern was small, acute, crossover, pilot, or repeated-measures work rather than large, preregistered, sport-specific trials.

The most practically relevant finding is not that one breathing method is clearly superior. Rather, the map shows that different breathing strategies are being used for different purposes and should be evaluated separately. Hyperventilation studies tend to target high-intensity power, sprint, blood-gas, or fatigue-related outcomes. Paced breathing is more often directed toward reaction time, recovery, heart-rate variability, stress, or anxiety. Hypoventilation and breath-holding studies are concentrated in repeated-sprint, swimming, judo, freediving, or oxygen-tolerance contexts. Pranayama/yogic studies often examine pulmonary, reaction-time, or broader performance-related outcomes.

The exploratory effect-size summaries are consistent with this interpretation. Hyperventilation showed a small, borderline pooled effect on power-related outcomes, but confidence intervals included a null effect. Paced breathing did not show a clear pooled benefit for reaction outcomes and showed moderate heterogeneity. These analyses are useful for identifying candidate domains for future systematic review, but they are not sufficient for definitive recommendations.

4.2 Comparison with existing literature

This review extends, rather than replaces, the existing syntheses. Laborde and colleagues conducted a systematic review and meta-analysis of breathing techniques on physical sport performance and found that longer-term slow-paced breathing and breath-holding interventions were associated with improved performance, whereas short-term slow-paced breathing, breath holding, and voluntary hyperventilation were not clearly related to performance (Laborde et al., 2024). They also emphasized high risk of bias in a substantial proportion of studies, limited evidence for less-studied breathing techniques, heterogeneous protocols, and the need for standardized intervention procedures. The present scoping review reaches a compatible conclusion from a broader mapping perspective: the field is promising, but technique labels, protocols, comparators, outcomes, and feasibility reporting remain too inconsistent for a single practical recommendation.

Migliaccio and colleagues offered a narrative synthesis of respiratory rate and sport performance, arguing that slow voluntary breathing may support cardiovascular regulation, stress reduction, concentration, and recovery, while fast voluntary breathing may be relevant for activation but can also produce dizziness, anxiety-like symptoms, or other undesirable responses if poorly controlled (Migliaccio et al., 2023). The present review adds a structured evidence map to that physiological and psychological rationale. It shows where those mechanisms have actually been tested in sport-relevant studies, which cohorts and sports have been studied, which outcome clusters are beginning to accumulate, and where the evidence is still thin.

The main added contribution of this review is therefore practical and methodological. Before this

map, the literature already suggested that breathing rate and breathing pattern could influence sport performance through autonomic, respiratory, attentional, and acid-base pathways. What is clearer now is that empirical sport research has developed unevenly: some techniques have entered athlete practice through coaching traditions and local experimentation before being tested with consistent protocols. The next step is not to dismiss those practices, but to test them rigorously, define the active breathing components, compare them with credible control conditions, and report feasibility outcomes so that promising routines can be separated from attractive but unsupported claims.

4.3 Implications for practice

Practitioners should avoid treating “breathwork” as a single intervention class. Acute hyperventilation, slow-paced breathing, hook breathing, hypoventilation, and pranayama differ in physiological targets, risk profile, and likely performance relevance. Overall, breathwork appears promising as a low-cost and adaptable family of practices, particularly when the technique is matched to the intended mechanism. The mapped evidence suggests that some methods may be promising in narrowly matched contexts, such as hyperventilation before or between high-intensity bouts, paced breathing for psychophysiological regulation, and hypoventilation in repeated-sprint or aquatic contexts. However, practical use should remain conservative because many studies are small, short-term, and incompletely reported.

4.4 Implications for research

Future studies should define the breathing intervention precisely, including respiratory rate, depth, timing relative to exercise, session frequency, total exposure, supervision, and safety monitoring. Trials should select outcomes that match the hypothesized mechanism, report adherence and dropout, and preregister analysis plans. Sport-specific studies should distinguish elite, competitive, recreational, and non-athlete populations. New breathing techniques that emerge empirically in sport settings should be treated as testable interventions rather than informal routines: they need clear protocols, plausible mechanisms, credible comparators, feasibility reporting, and adverse-event monitoring. The field would also benefit from a shared taxonomy separating ventilatory manipulation, autonomic regulation, respiratory-muscle practice, apnea/hypoventilation exposure, and meditative or yogic breathing practices.

4.5 Limitations

This review has several limitations. The search was based on PubMed and citation backtracking, so relevant studies indexed elsewhere may have been missed. The included literature was heterogeneous in design, participant level, breathing technique, comparator condition, dose, timing, and outcome selection. Many studies had small samples, short exposure periods, or acute crossover designs, which limits inference about sustained training effects and sport-specific performance transfer. Reporting of adherence, dropout, adverse events, race/ethnicity, and some baseline characteristics was incomplete across the corpus.

The effect-size analyses also have limitations. They rely on extracted group means and dispersions, combine only small clusters, and pool outcomes that are conceptually related but not identical. As a result, the quantitative summaries should be interpreted as exploratory signals within a scoping review, not as conclusive evidence of efficacy.

5 Conclusions

The evidence map shows a rapidly developing but heterogeneous literature on breathing strategies in sport and performance settings. The current evidence is strongest as a map of concepts, contexts, and gaps. Breathwork appears promising, especially when a technique is linked to a plausible physiological or psychological mechanism, but the field still needs rigorous testing of sport-derived breathing routines that have often developed unsystematically in applied settings. Hyperventilation may have a small acute signal for power-related outcomes, while paced-breathing effects on reaction outcomes remain uncertain. Feasibility and safety reporting are major weaknesses across the corpus. Future work should use clearer taxonomies, complete reporting, feasibility outcomes, and sport-specific, adequately powered comparative designs before strong practical recommendations are made.

Declarations

Funding: No external funding was declared.

Conflicts of interest: The authors declare no conflicts of interest.

Data availability: Extracted data, analysis code, tables, and figures are available from the study repository.

Author contributions: Author contributions will be completed before submission.

A PRISMA-ScR Checklist Crosswalk

Table 8: PRISMA-ScR reporting checklist crosswalk for this manuscript.

Item	Reporting element	Location in manuscript
Title	Identifies the report as a scoping review	Title page
Abstract	Structured summary including background, objectives, eligibility, sources, charting, results, conclusions	Abstract
Rationale	Rationale for mapping the evidence	Introduction
Objectives	Review question and objectives	Introduction
Protocol and registration	Protocol status reported	Methods
Eligibility criteria	Population, concept, context, source characteristics	Methods
Information sources	PubMed and citation backtracking reported	Methods
Search	Search source count reported; exact string/date unavailable	Methods; limitations
Selection of sources	Screening stages and counts reported	Methods; Figure 1
Data charting process	Extraction, cleaning, and charting variables reported	Methods

Item	Reporting element	Location in manuscript
Data items	Bibliometric, demographic, intervention, outcome, feasibility, and safety variables reported	Methods
Critical appraisal	Not performed as a formal quality appraisal; data completeness appraised descriptively	Methods; limitations
Synthesis	Narrative mapping plus exploratory effect-size summaries	Methods; Results
Selection results	Counts at each screening stage	Results; Figure 1
Characteristics	Included-source characteristics	Results; Table 7
Critical appraisal results	Data-quality issues summarized where relevant	Results; limitations
Individual-source results	Study-level characteristics and outcome clusters summarized	Results; appendices
Synthesis of results	Evidence map, plots, tables, and exploratory syntheses	Results
Summary of evidence	Main findings linked to objectives	Discussion
Limitations	Review and evidence-base limitations	Discussion
Conclusions	Interpretation and next steps	Conclusions
Funding	Funding statement placeholder included	Declarations

B Referenced Included Sources

References

- A, S., H, N., & CM, C. (2014). Hyperventilation as a strategy for improved repeated sprint performance. *Journal of strength and conditioning research*, 28(4), 1119–26. <https://pubmed.ncbi.nlm.nih.gov/23838981/>
- A, S., H, N., & CM, C. (2015). Hyperventilation-induced respiratory alkalosis falls short of countering fatigue during repeated maximal isokinetic contractions. *European journal of applied physiology*, 115(7), 1453–65. <https://pubmed.ncbi.nlm.nih.gov/25688041/>
- A, S., H, N., & CM, C. (2018). Effects of hyperventilation on repeated pedaling sprint performance: Short vs. long intervention duration. *Journal of strength and conditioning research*, 32(1), 170–180. <https://pubmed.ncbi.nlm.nih.gov/28135216/>
- Arksey, H., & O'Malley, L. (2005). Scoping studies: Towards a methodological framework. *International Journal of Social Research Methodology*, 8(1), 19–32. <https://doi.org/10.1080/1364557032000119616>
- BA, S., DA, J., & KW, R. (2004). An evaluation of standardizing target ventilation for eucapnic voluntary hyperventilation using fev1. *The Journal of asthma : official journal of the Association for the Care of Asthma*, 41(7), 745–9. <https://pubmed.ncbi.nlm.nih.gov/15584634/>

- Buchanan, T. L., & Janelle, C. M. (2021). Fast breathing facilitates reaction time and movement time of a memory-guided force pulse. *Human Movement Science*, 76, 102762. <https://doi.org/10.1016/j.humov.2021.102762>
- CS, H., R, B., & MN, K. (2017). Yogic breathing practices improve lung functions of competitive young swimmers. *Journal of Ayurveda and integrative medicine*, 8(2), 99–104. <https://pubmed.ncbi.nlm.nih.gov/28601355/>
- D, M., P, B., V, B., GJ, G., & JD, V. (2022). Does wim hof method improve breathing economy during exercise? *Journal of clinical medicine*, 11(8). <https://pubmed.ncbi.nlm.nih.gov/35456308/>
- EY, F.-B., LF, R.-S., S, H.-M., JM, L.-W., & AG, M.-M. (2026). Slow paced breathing and power posing: A pre-competitive ritual during an american football season. *Frontiers in sports and active living*, 8, 1800382. <https://pubmed.ncbi.nlm.nih.gov/42256048/>
- F, C., B, G., M, E., C, P., M, E., A, S., M, G., C, L. C., P, L., M, C., C, H., D, L., & C, T. (2026). Exploring the influence of coherent breathing on psychophysiological stress during a simulated 3-day 400-m race. *Journal of strength and conditioning research*, 40(5), e460–e471. <https://pubmed.ncbi.nlm.nih.gov/41359888/>
- F, D., P, M., & C, P. (2000). Evidence for an inadequate hyperventilation inducing arterial hypoxemia at submaximal exercise in all highly trained endurance athletes. *Medicine and science in sports and exercise*, 32(5), 926–32. <https://pubmed.ncbi.nlm.nih.gov/10795782/>
- HH, Y., S, U., F, K., NF, K., C, G., S, Ā.-z., & Ā. (2025). Voluntary hyperventilation-induced hypocapnia enhances wingate test performance without altering energy system contributions. *BMC sports science, medicine and rehabilitation*, 17(1), 167. <https://pubmed.ncbi.nlm.nih.gov/40604959/>
- IE, D., & I, E. (2025). Influence of breathing strategies on maximal strength output and hemodynamic parameters during bench press exercise. *BMC sports science, medicine and rehabilitation*, 18(1), 22. <https://pubmed.ncbi.nlm.nih.gov/41366700/>
- L, F., A, P., V, H., NS, L., O, O., N, P., & TL, B. (2026). The effect of pre-exercise hyperventilation on repeated high-intensity inclined sprint performance. *Experimental physiology*. <https://pubmed.ncbi.nlm.nih.gov/42185035/>
- Laborde, S., Zammit, N., Iskra, M., Mosley, E., Borges, U., Allen, M. S., & Javelle, F. (2024). The influence of breathing techniques on physical sport performance: A systematic review and meta-analysis. *International Review of Sport and Exercise Psychology*, 17(2), 1222–1277. <https://doi.org/10.1080/1750984X.2022.2145573>
- Levac, D., Colquhoun, H., & O'Brien, K. K. (2010). Scoping studies: Advancing the methodology. *Implementation Science*, 5, 69. <https://doi.org/10.1186/1748-5908-5-69>
- M, A. A. B., Y, O., & X, W. (2024). Effects of a 6-week repeated-sprint training with voluntary hypoventilation at low and high lung volume on repeated-sprint ability in female soccer players. *International journal of sports physiology and performance*, 19(5), 463–470. <https://pubmed.ncbi.nlm.nih.gov/38412852/>
- M, I., S, L., T, P., C, S., E, W., M, R., & L, V. (2026). Specifying the cardio-respiratory patterns during fast-paced breathing. *Psychophysiology*, 63(5), e70305. <https://pubmed.ncbi.nlm.nih.gov/42057338/>

- M, K., & GR, A. (2025). Box breathing or six breaths per minute: Which strategy improves athletes post-hiit cardiovascular recovery? *PloS one*, 20(11), e0336615. <https://pubmed.ncbi.nlm.nih.gov/41248139/>
- M, M., E, M., & A, M. (2014). The influence of different modes of ventilation on standing balance of athletes. *Asian journal of sports medicine*, 5(3), e22767. <https://pubmed.ncbi.nlm.nih.gov/25520763/>
- M, S., R, M., O, Å., J, K., A, Å., S, K.-N., & D, J. (2024). Influence of the breathing pattern on the pulmonary function of endurance-trained athletes. *Scientific reports*, 14(1), 1113. <https://pubmed.ncbi.nlm.nih.gov/38212427/>
- M, Y., S, L., S, A., U, B., F, D., & E, M. (2024). Influence of respiratory frequency of slow-paced breathing on vagally-mediated heart rate variability. *Applied psychophysiology and biofeedback*, 49(1), 133–143. <https://pubmed.ncbi.nlm.nih.gov/38063977/>
- Migliaccio, G. M., Russo, L., Maric, M., & Padulo, J. (2023). Sports performance and breathing rate: What is the connection? a narrative review on breathing strategies. *Sports*, 11(5), 103. <https://doi.org/10.3390/sports11050103>
- N, F., H, B., & J, K. (2025). A semi-randomised control trial assessing psychophysiological effects of breathwork and cold immersion. *Scientific reports*, 15(1), 43879. <https://pubmed.ncbi.nlm.nih.gov/41388053/>
- NS, K., LM, R., & CA, F. (2026). The acute effect of a diaphragmatic breathing warm-up on maximal deadlift performance. *Journal of strength and conditioning research*. <https://pubmed.ncbi.nlm.nih.gov/42155066/>
- P, B., V, B., R, M., D, M., GJ, G., & J, S. (2021). Impact of a breathing intervention on engagement of abdominal, thoracic, and subclavian musculature during exercise, a randomized trial. *Journal of clinical medicine*, 10(16). <https://pubmed.ncbi.nlm.nih.gov/34441810/>
- P, L., & S, N. (2005). Studies on inspiratory and expiratory glossopharyngeal breathing in breath-hold divers employing magnetic resonance imaging and spirometry. *European journal of applied physiology*, 94(5), 646–51. <https://pubmed.ncbi.nlm.nih.gov/15942772/>
- P, R., B, W., S, L., & R, C. (2026). No evidence for the efficacy of slow-paced breathing as a recovery strategy after sprint interval training. *International journal of sports physiology and performance*, 21(2), 302–311. <https://pubmed.ncbi.nlm.nih.gov/41569822/>
- Peters, M. D. J., Marnie, C., Tricco, A. C., Pollock, D., Munn, Z., Alexander, L., McInerney, P., Godfrey, C. M., & Khalil, H. (2020). Updated methodological guidance for the conduct of scoping reviews. *JBI Evidence Synthesis*, 18(10), 2119–2126. <https://doi.org/10.11124/JBIES-20-00167>
- Q, M., P, V., L, M., G, L., G, M., & M, N. (2024). Psychophysiological effects of slow-paced breathing on adolescent swimmers' subjective performance, recovery states, and control perception. *Journal of functional morphology and kinesiology*, 9(1). <https://pubmed.ncbi.nlm.nih.gov/38390923/>
- SS, V., B, A., Y, S., & R, M. (2021). Effect of yoga on performance and physical fitness in cricket bowlers. *International journal of yoga therapy*, 31(1). <https://pubmed.ncbi.nlm.nih.gov/34551075/>
- T, C., von Roten F, C., B, K., & K, G. (2021). Acute effects of the wim hof breathing method on repeated sprint ability: A pilot study. *Frontiers in sports and active living*, 3, 700757. <https://pubmed.ncbi.nlm.nih.gov/34514386/>

- Tricco, A. C., Lillie, E., Zarin, W., O'Brien, K. K., Colquhoun, H., Levac, D., Moher, D., Peters, M. D. J., Horsley, T., Weeks, L., Hempel, S., Akl, E. A., Chang, C., McGowan, J., Stewart, L., Hartling, L., Aldcroft, A., Wilson, M. G., Garritty, C., ... Straus, S. E. (2018). Prisma extension for scoping reviews (prisma-scr): Checklist and explanation. *Annals of Internal Medicine*, 169(7), 467–473. <https://doi.org/10.7326/M18-0850>
- V, S., AG, T., & E, K. (2015). Changes in respiratory parameters and fin-swimming performance following a 16-week training period with intermittent breath holding. *Journal of human kinetics*, 49, 89–98. <https://pubmed.ncbi.nlm.nih.gov/26839609/>
- X, W., C, F., SP, D., F, B., P, R., P, C., JV, B., S, V., AF, G., G, M., O, D., & A, P. (2024). Hypoventilation training including maximal end-expiratory breath holding improves the ability to repeat high-intensity efforts in elite judo athletes. *Frontiers in physiology*, 15, 1441696. <https://pubmed.ncbi.nlm.nih.gov/39397858/>
- Y, L., J, W., Y, L., D, L., Y, X., & Y, L. (2024). The impact of starting positions and breathing rhythms on cardiopulmonary stress and post-exercise oxygen consumption after high-intensity metabolic training: A randomized crossover prospective study. *Healthcare (Basel, Switzerland)*, 12(18). <https://pubmed.ncbi.nlm.nih.gov/39337230/>
- Y, T., M, Y., K, S., T, I., E, S., K, I., F, T., & K, T. (2026). Focused attention meditation as a pre-exercise strategy for reducing anxiety in speed skaters. *Sensors (Basel, Switzerland)*, 26(2). <https://pubmed.ncbi.nlm.nih.gov/41600272/>